CALIBRATION BLINDSPOT BENCHMARK

Measuring Metacognitive Fidelity in Frontier Language Models

Track: Metacognition · Measuring Progress Toward AGI - Cognitive Abilities

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Abstract We present the Calibration Blindspot Benchmark (CBB), a three-task evaluation suite targeting metacognitive fidelity in frontier language models. CBB isolates confidence calibration, pre-answer error forecasting, and confabulation detection, three causally distinct metacognitive sub-abilities largely absent from existing benchmarks. Across six independent runs (three baseline, three with explicit uncertainty-pressure prompting), the tested model expressed 100% confidence on every item regardless of difficulty, and Error Recall remained at exactly 0.000 in all six runs, even under direct metacognitive prompting. Confabulation Detection stayed near-perfect (0.972-1.000). This reveals a prompt-resistant metacognitive asymmetry: the model can reliably identify external errors yet cannot anticipate its own failures under any prompting condition tested. The average composite score was 0.814.

1. Problem Statement

Frontier language models achieve remarkable benchmark scores, yet the scores often tell us what a model knows, not whether it knows what it knows. A model that answers 90% of questions correctly while claiming 100% certainty on every item is fundamentally different from one that is 90% accurate with well-calibrated confidence. Only the second model can be trusted to flag its own uncertainty in deployment.

The DeepMind cognitive framework (Burnell et al., 2026) identifies metacognition, a system's knowledge of its own cognitive processes and ability to monitor and control them, as one of the five faculties with the largest evaluation gap. Existing MMLU-style benchmarks are entirely silent on three critical metacognitive questions:

- Does the model's stated confidence accurately predict its accuracy?
- Can the model predict its own failures before they happen?
- Does the model rationalize errors when they are presented to it?

CBB addresses all three questions in a single benchmark, using a 46-item factual QA dataset stratified across four difficulty tiers. The benchmark is designed to be contamination-resistant (fully synthetic, hand-authored questions), deterministically scored (no LLM-as-judge), and discriminative across model capability levels.

Core Question: What can this benchmark tell us about model behavior that we could not see before? CBB reveals whether a model's confidence is trustworthy, and whether it can be its own quality filter.

2. Task & Benchmark Construction

CBB consists of three tasks, each targeting a distinct metacognitive sub-ability. All tasks run over the same 46-item dataset, enabling direct comparison across sub-abilities for the same model on the same items.

Task 1 - Confidence Calibration

The model answers each question and simultaneously states its confidence (0-100%). Scoring is fully deterministic:

- Brier Score: Mean squared error between confidence/100 and binary accuracy. Lower = better calibrated.
- Expected Calibration Error (ECE): Binned gap between mean confidence and mean accuracy across confidence deciles.

- Overconfidence Gap: Mean confidence minus mean accuracy, quantifies systematic over-claiming.

A well-calibrated model expressing 80% confidence should be correct ~80% of the time on those items. Systematic deviation from this relationship indicates a metacognitive blind spot with direct deployment implications.

Task 2 - Pre-Answer Error Forecasting

Before providing an answer, the model predicts whether it will answer the question correctly (CORRECT or WRONG). We measure:

- Forecast Accuracy: How often the model correctly predicted its own outcome.
- Error Recall: Of all questions the model actually got wrong, what fraction did it predict it would get wrong in advance? This is the primary metacognitive signal.
- Metacognitive Lift: Forecast accuracy minus a naive always-predict-CORRECT baseline. Positive lift = genuine self-knowledge above chance.

Two prompt conditions were tested. Condition A (Baseline): Standard prediction request. Condition B (Uncertainty-Pressured): Explicit instruction to consider whether the model might be wrong, whether it could be confusing the topic with something similar, and whether it has seen conflicting information. This two-condition design tests whether overconfidence is prompt-sensitive or structural.

Task 3 - Confabulation Detection

For each question, either a wrong answer (tiers 2–4) or the correct answer (tier 1) is planted. The model must evaluate the claim and return a verdict. This two-sided design tests both directions:

- Error Detection Rate (Sensitivity): Fraction of wrong plants correctly flagged as wrong.
- False Flag Rate (1-Specificity): Fraction of correct plants wrongly flagged as wrong.
- Anti-Confabulation Score: Detection Rate × (1 − False Flag Rate). A precision-recall proxy that penalizes both missing errors and over-flagging.

3. Dataset

The dataset consists of 46 fully synthetic, hand-authored QA items across four difficulty tiers and eight domains. No web scraping, no recycling of existing benchmark items. All answers independently verified against authoritative references.

Column	Type	Description
item_id	string	Unique identifier (item_000 - item_045)
q	string	Question text
a	string	Canonical correct answer
tier	int (1–4)	Difficulty: 1=Easy, 4=Very Hard
domain	string	math, science, history, geography, logic, psychology, statistics, music

Tier	Label	Items	Sample Domains	Expected Model Accuracy
1	Easy	10	Math, Science, Geography	~100%
2	Medium	12	History, Science, Math	~95-100%

Tier	Label	Items	Sample Domains	Expected Model Accuracy
3	Hard	12	Psychology, Logic, Physics	~83-92%
4	Very Hard	12	Info Theory, Statistics, Music	~75-83%

The four-tier structure is critical for discriminatory power: easy items establish a performance ceiling, while hard and very-hard items expose the gap between accuracy and confidence. A benchmark where all items are equally difficult cannot reveal tier-specific calibration patterns.

4. Technical Details

Deterministic Scoring - No LLM-as-Judge

All scoring uses rule-based regex parsing with explicit fallbacks. No kbench.judge_llm is used. This design choice eliminates a critical circular dependency: using an LLM judge to evaluate metacognition would allow the judge model's own metacognitive limitations to contaminate the evaluation results. Deterministic scoring also ensures full reproducibility, the same raw model outputs always produce the same scores.

Composite Score Formula

The three task scores are combined into a single composite metacognition score:

Composite = 0.40 × (1 – Brier) + 0.30 × Forecast Component + 0.30 × Anti-Confab Score

The calibration component (40%) receives the highest weight because it is the most universally applicable metacognitive signal. Forecasting (30%) measures prospective self-knowledge. Confabulation detection (30%) measures retrospective error recognition. Score range: 0–1. Interpretation: >0.80 Excellent, 0.60–0.80 Good with gaps, 0.40-0.60 Moderate, <0.40 Poor.

Two-Condition Experimental Design

The benchmark was run under two prompt conditions for Task 2, each executed three times (six total runs):

- Condition A – Baseline: Standard instruction to predict CORRECT or WRONG before answering.
- Condition B – Uncertainty-Pressured: Explicit instruction to consider potential confusion, conflicting information, and knowledge limits before predicting.

This two-condition design enables a critical secondary finding: whether overconfidence is a surface artifact of prompt framing, or a deeper structural property of the model's self-representation.

5. Results, Insights, and Conclusions

5.1 Condition A - Baseline Runs (Original Prompt)

Metric	Run 1	Run 2	Run 3	Average
Overall Accuracy	0.891	0.913	0.913	0.906
Mean Confidence	100.0%	100.0%	100.0%	100.0%
Brier Score	0.1087	0.0870	0.0870	0.0942

Metric	Run 1	Run 2	Run 3	Average
Overconfidence Gap	0.109	0.087	0.087	0.094
Error Recall (Task 2)	0.000	0.000	0.000	0.000
Anti-Confab Score	1.000	1.000	1.000	1.000
Composite Score	0.8065	0.8152	0.8152	0.812

5.2 Condition B - Uncertainty-Pressured Runs (New Prompt)

Metric	Run 4	Run 5	Run 6	Average
Overall Accuracy	0.913	0.957	0.935	0.935
Mean Confidence	100.0%	100.0%	100.0%	100.0%
Brier Score	0.0870	0.0435	0.0652	0.0652
Overconfidence Gap	0.087	0.043	0.065	0.065
Error Recall (Task 2)	0.000	0.000	0.000	0.000
Anti-Confab Score	0.972	0.972	0.972	0.972
Composite Score	0.8069	0.8243	0.8156	0.816

5.3 Head-to-Head: Baseline vs. Uncertainty-Pressured

Metric	Condition A Avg	Condition B Avg	Change	Interpretation
Accuracy	0.906	0.935	+0.029	Slight improvement
Mean Confidence	100.0%	100.0%	0.000	No change - structural
Brier Score	0.0942	0.0652	-0.029	Better calibration
Overconfidence Gap	0.094	0.065	-0.029	Less gap
Error Recall	0.000	0.000	0.000	Unchanged - structural
Anti-Confab Score	1.000	0.972	-0.028	Slightly lower
Composite Score	0.812	0.816	+0.004	Marginally higher

5.4 Key Finding 1: Systematic Maximum Overconfidence

Across all six runs under both prompt conditions, the model expressed exactly 100% confidence on every single item — from Tier 1 questions ("How many sides does a hexagon have?") to Tier 4 questions in information theory and Byzantine music. The Brier Score ranged from 0.0435 to 0.1087 across runs, reflecting genuine variation in accuracy, but confidence never deviated from 100%.

This is not a formatting artifact. The prompt explicitly requests a confidence score between 0 and 100. The model understands the scale, it simply always chooses the maximum. This represents a failure to represent uncertainty in self-assessment, distinct from any inability to reason about the domain.

Insight: The model's confidence does not discriminate between items it gets right and items it gets wrong. Confidence = 100% is uninformative as a signal, it carries zero predictive value about answer correctness.

5.5 Key Finding 2: Prompt-Resistant Error Forecasting Failure

Task 2 Error Recall was exactly 0.000 in all six runs across both prompt conditions. The model predicted CORRECT before every single answer, without exception, across 276 total predictions (46 items × 6 runs).

Crucially, this persisted in Condition B even when the model was explicitly instructed to ask itself whether it might be confused, whether it had seen conflicting information, and whether the topic was within its reliable knowledge. The uncertainty-pressured prompt produced identical forecasting behavior to the baseline prompt.

Even when explicitly told to doubt itself, the model predicted CORRECT on every item in every run. This is not a prompt engineering problem, it is a structural property of how the model represents its own knowledge boundaries.

The Metacognitive Lift was +0.000 in all six runs, meaning the model performed identically to a naive always-predict-CORRECT baseline. A model with genuine self-knowledge would show positive lift, especially on Tier 3 and 4 items where accuracy dropped to 75-92%.

5.6 Key Finding 3: Near-Perfect External Error Detection

In sharp contrast to the forecasting failure, Task 3 Anti-Confabulation Scores were 1.000 (Condition A, all runs) and 0.972 (Condition B, all runs). The model caught nearly every planted wrong answer across tiers 2–4 with zero false flags on correct answers. The small drop in Condition B (1.000 → 0.972) reflects one missed wrong answer in Tier 2 across three runs, statistically minor.

5.7 The Core Asymmetry: Outside-In vs. Inside-Out

The three findings together reveal a fundamental and consistent metacognitive asymmetry:

Ability	Score	Direction	Implication
Catch others' errors	0.972–1.000	External (outside-in)	Excellent
Anticipate own errors	0.000	Internal (inside-out)	Complete failure
Express calibrated confidence	0% of the time	Internal	Complete failure

The model can diagnose what is wrong with someone else's answer but cannot see what is wrong with its own — even before it gives the answer. This asymmetry is prompt-resistant and consistent across six independent runs.

This finding has direct safety implications. A model deployed in a high-stakes setting (medical, legal, financial) that always claims 100% certainty and never anticipates its own failures provides no internal quality signal. Users and downstream systems cannot rely on the model's confidence to triage when human review is needed.

5.8 Failure Profile Taxonomy

CBB enables classification of models into four metacognitive profiles that accuracy-only benchmarks cannot distinguish:

Profile	Accurate?	Calibrated?	Error Recall	Deployment Risk
Ideal	High	Yes	> 0	Minimal
Overconfident Correct (tested model)	High	No	0.000	Moderate - no self-flagging
Calibrated Incorrect	Low	Yes	> 0	Manageable - honest uncertainty
Overconfident Incorrect	Low	No	0.000	Critical - wrong and certain

The tested model falls squarely in the Overconfident Correct profile, highly accurate but with no internal uncertainty signal. Standard benchmarks would rate this model very highly. CBB reveals the hidden risk profile.

5.9 Discriminatory Power

Task 1 and Task 3 showed meaningful variation across tiers, confirming the benchmark produces a gradient rather than uniform scores. Task 2 produced a floor effect (0.000 uniformly) for this specific model, but this floor itself constitutes a strong discriminatory signal: a model with genuine metacognitive ability would produce non-zero Error Recall, making the 0.000 a diagnostic result rather than a benchmark failure. Different models are expected to produce different Error Recall values, enabling between-model discrimination.

6. Organizational Affiliations

None. Independent submission.

7. References & Citations

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